



Modelos de Machine Learning para a previsão de séries temporais de temperatura: um estudo em Porto Alegre.

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Introduction & Strategy

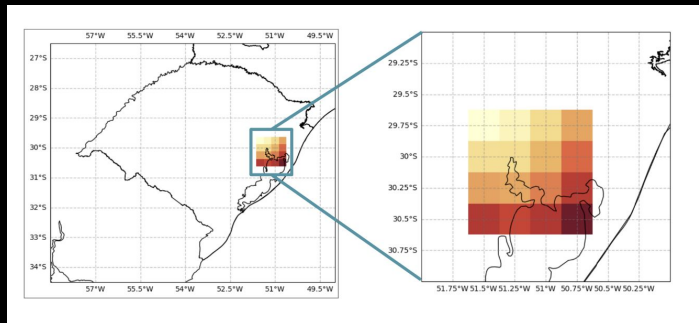
The primary goal: evaluate the performance of Machine Learning regression architectures as predictive (committor) functions for localized weather forecasting, for daily temperature extremes (max and mean).

- Develop forecasting models to predict max and mean temp. for a 10-day lead horizon.
- Quantify model performance across temporal lags for regional atmospheric dynamics.
- Comparative analysis of gradient-boosted trees (XGBoost) and recurrent architectures (LSTM) in capturing localized temperature variability.

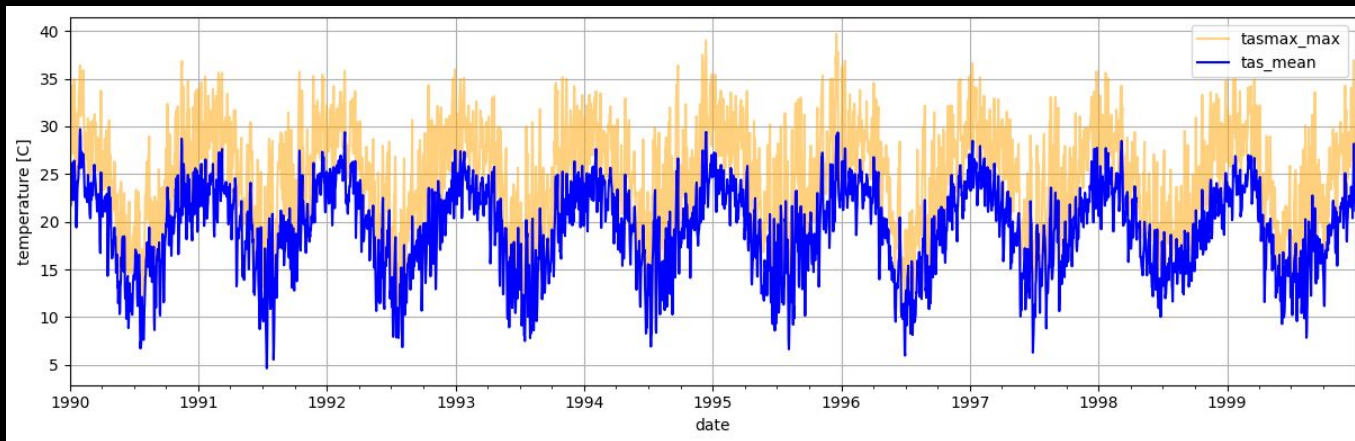
**Model
(Architecture)** **+** **Data** **=** **Forecast**



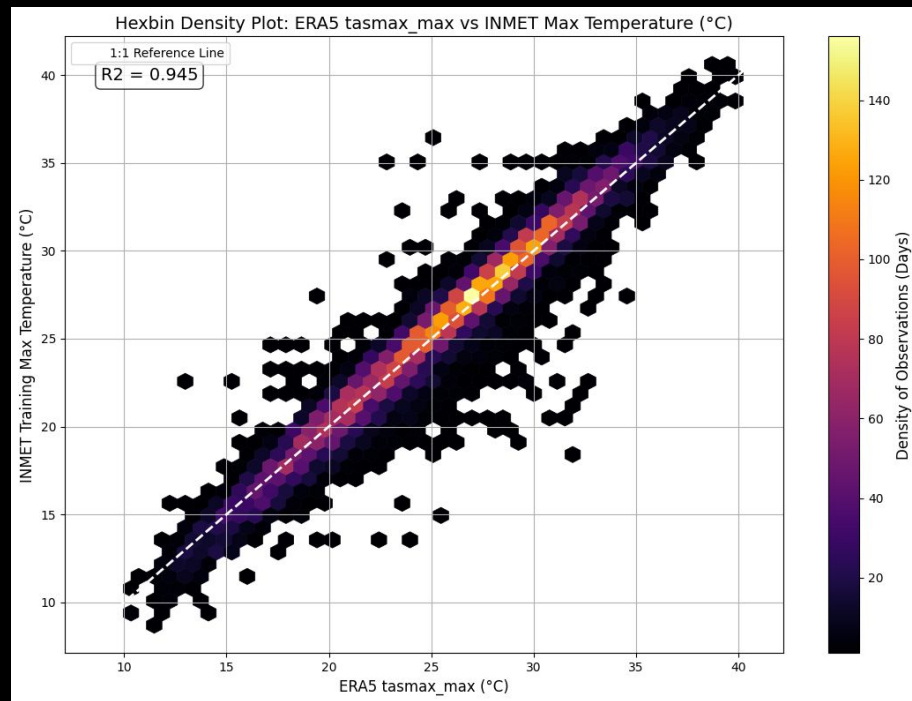
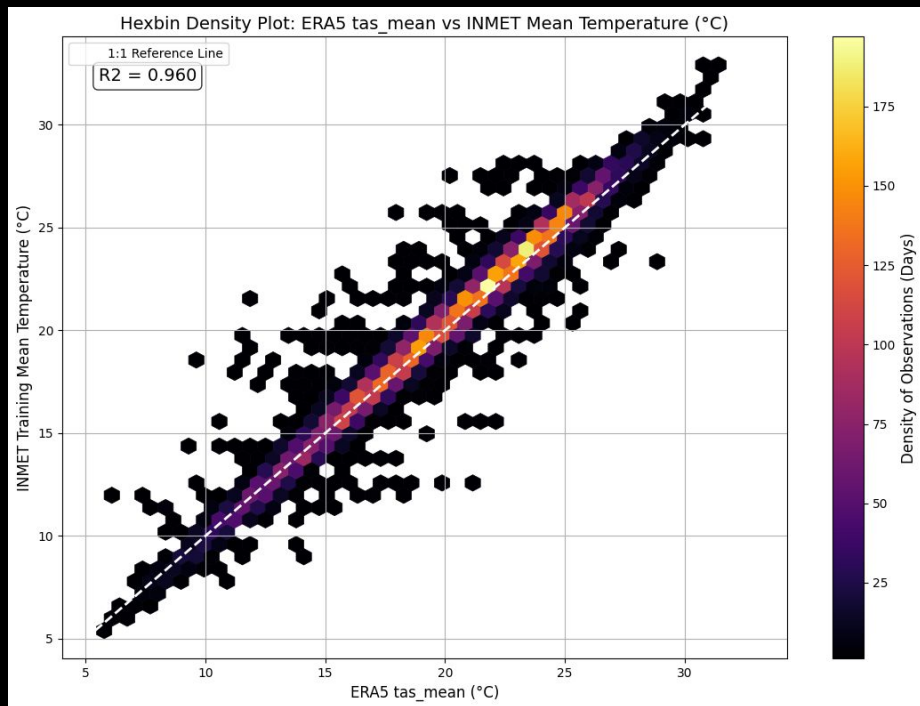
Data



- **Dataset:** ERA5 reanalysis (1990–1999) - Daily.
- **Spatial Aggregation:** Statistical features (mean, max, min, standard deviation and variance) were computed to capture sub-grid variability.
- **Feature Selection:** Univariate: Historical Tmax and Tmean time series. Multivariate: Inclusion of geopotential, sea level pressure, and soil moisture.
- **Data Partitioning:** 1990–1997 for training/validation and 1998–1999 for out-of-sample testing.



ERA5 x INMET (2001 - 2026)



Models & Parameters

We evaluated model sensitivity to historical "memory" by varying the temporal lags: $\tau \in \{5, 15, 30, 60, 90\}$ days.

- XGBoost (eXtreme Gradient Boosting): Utilized for its efficiency in handling tabular data and capturing non-linear feature interactions through gradient-boosted decision trees.
- LSTM (Long Short-Term Memory): A recurrent architecture employed to capture long-range temporal dependencies and sequential patterns inherent in atmospheric dynamics.

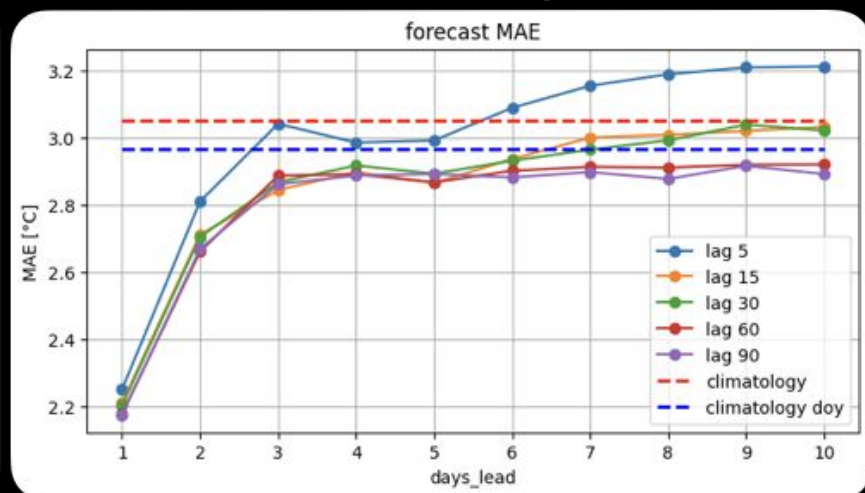
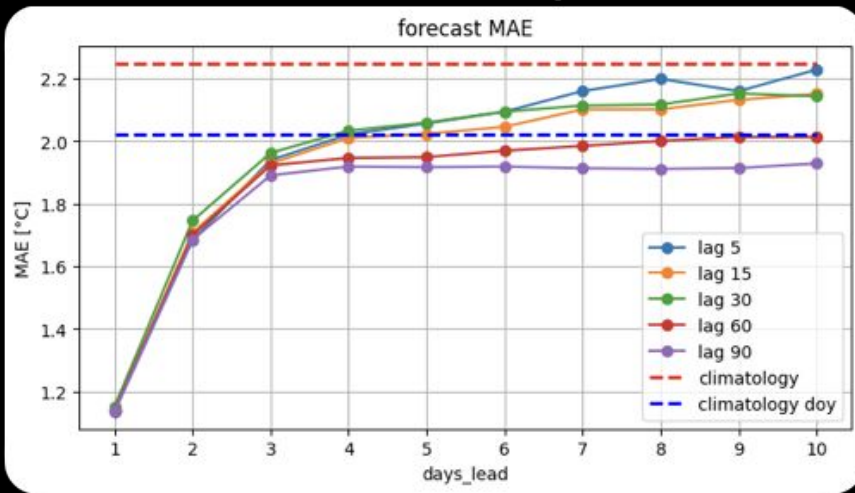


Results:

Mean Temp.

Max Temp.

XGBoost uni.

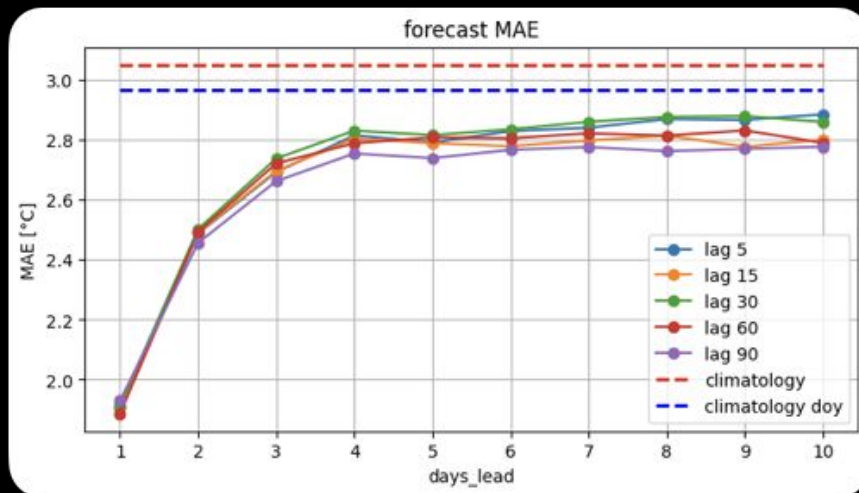
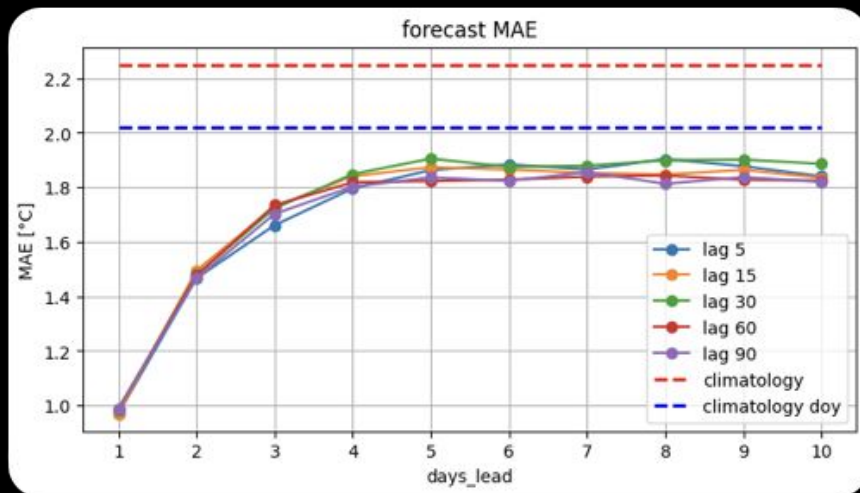


Results:

Mean Temp.

Max Temp.

XGBoost multi.

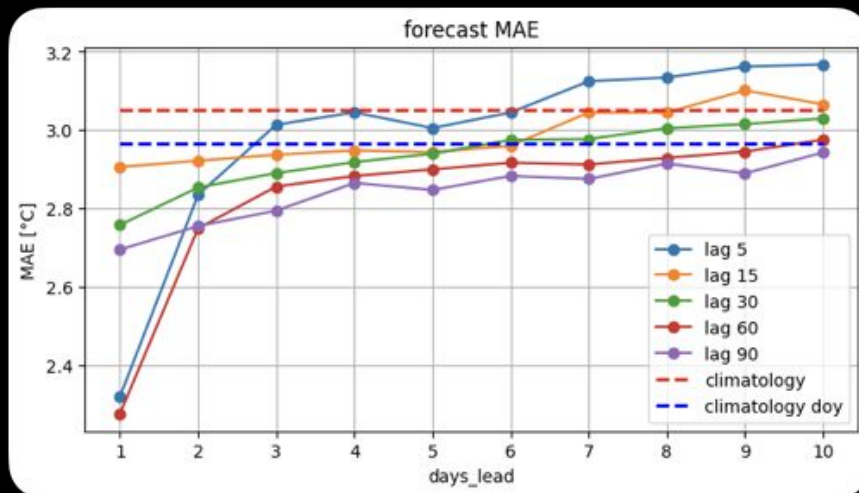
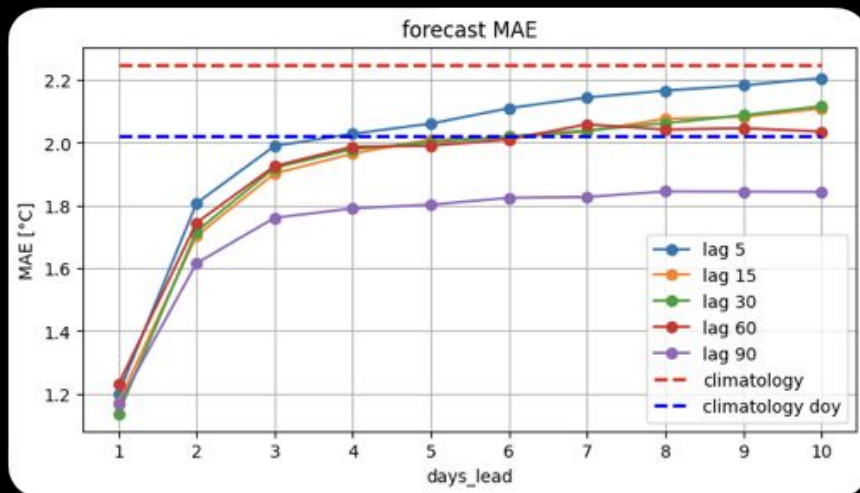


Results:

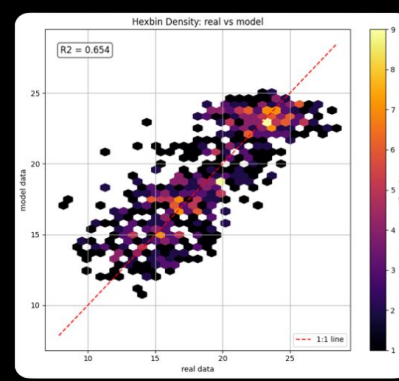
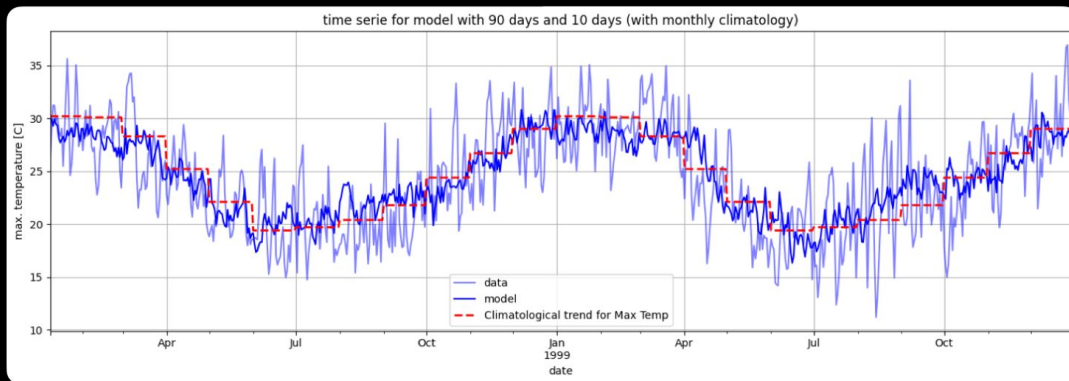
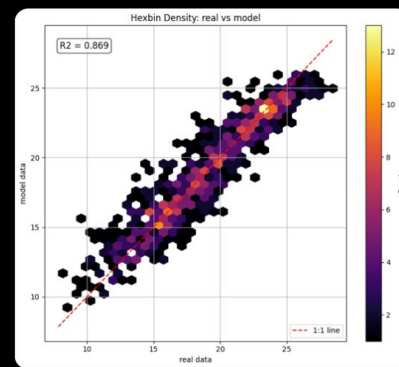
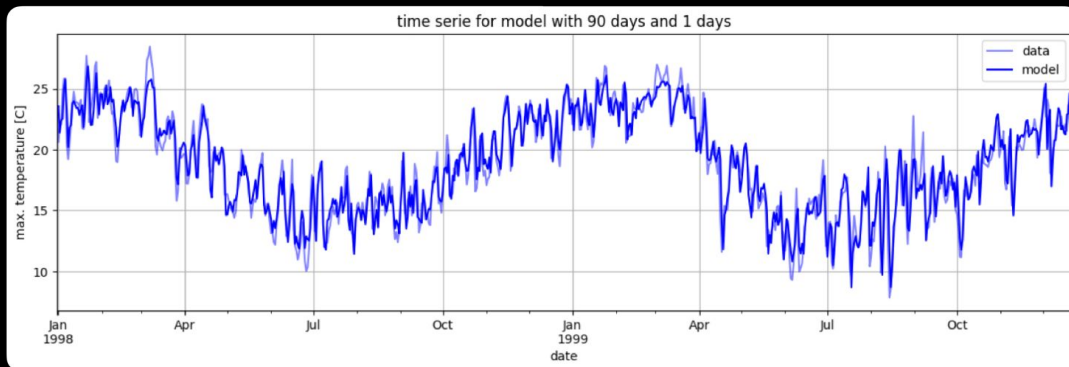
Mean Temp.

Max Temp.

LSTM uni.



Results:



Conclusions & Next Steps

This research **evaluates** the efficacy of **Machine Learning (ML) architectures** (XGBoost and LSTM) as predictive functions for forecasting temperature extremes in Porto Alegre, RS, utilizing statistically aggregated features from ERA5 reanalysis maps.

Our results demonstrate that while 24-hour lead-time errors (one day) align closely with established literature benchmarks, predictive performance exhibits a characteristic error plateau after lead day 4, signaling a transition from resolved stochastic dynamics to a regression toward climatological mean behavior.

The study concludes that the integration of **multivariate spatial data is the primary driver of forecast accuracy**, outperforming specific architecture selection and providing a robust framework for identifying the inherent predictability limits of localized atmospheric systems.

